| Name: | Student ID: |
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PROBABILITY PROBLEM SHEET

This sheet covers the lectures on probability, on interpreting experimental results, and on Bayesian inference. Question 1 practises the rules of probability, Question 2 error bars and significance tests, and Question 3 Bayesian updating. Each ends with a short Python part (`import numpy as np` is assumed). Question 4 is harder and optional: it does Bayesian updating for normal distributions and for linear regression.

**1**

(a) Events $A$ and $B$ have $\mathbb{P}(A) = 0.3$, $\mathbb{P}(B) = 0.5$ and $\mathbb{P}(A \cap B) = 0.1$. Find $\mathbb{P}(A \cup B)$, $\mathbb{P}(\neg A)$ and $\mathbb{P}(A \mid B)$. Are $A$ and $B$ independent? [4 marks]

$\overline{4}$

(b) The joint distribution of the weather $W$ and whether I carry an umbrella $U$ is

| | $U =$ yes | $U =$ no |
|---|---|---|
| $W =$ sun | 0.05 | 0.55 |
| $W =$ rain | 0.25 | 0.15 |

Find $\mathbb{P}(U = \text{yes})$, $\mathbb{P}(W = \text{rain} \mid U = \text{yes})$ and $\mathbb{P}(U = \text{yes} \mid W = \text{rain})$. Comment on the last two. [5 marks]

$\overline{5}$

(c) $X$ is the score on a fair dice. Find $\mathbb{E}\big[X\big]$, $\mathbb{E}\big[X^2\big]$ and $\mathrm{Var}(X)$. Use linearity of expectation to find $\mathbb{E}\big[2X+3\big]$, and find $\mathrm{Var}(2X+3)$. [4 marks]

$\frac{}{4}$

(d) $X$ is uniform on $(0,1)$ and $Y = -\log(X)/\lambda$ with $\lambda > 0$. Use the change of variables formula $f_Y(y) = f_X(x)|\mathrm{d}x/\mathrm{d}y|$ to find the density of $Y$. [4 marks]

$\frac{}{4}$

(e) In Python, draw $100000$ samples of $Y$ from part (d) with $\lambda = 2$. Check that their mean is close to $1/\lambda$, and estimate $\mathbb{P}(Y \leq 0.5)$ as the average of the indicator $[[Y \leq 0.5]]$. Compare with the exact value $1 - \mathrm{e}^{-\lambda/2}$. [3 marks]

$\frac{}{3}$

End of question 1

| (a) $\frac{}{4}$ | (b) $\frac{}{5}$ | (c) $\frac{}{4}$ | (d) $\frac{}{4}$ | (e) $\frac{}{3}$ | Total $\frac{}{20}$ |
|---|---|---|---|---|---|

**2**

(a) A classifier gets 1700 of 2000 test examples right. Estimate its accuracy and standard error, and write the result down sensibly. [3 marks]

$\overline{3}$

(b) A second classifier gets $0.866 \pm 0.008$ on a *different* test set of 2000 examples. Is it significantly better at the 5% level? [4 marks]

$\overline{4}$

(c) Now both classifiers are run on the *same* 2000 examples. The first is right and the second wrong on 20 examples; the second is right and the first wrong on 52; they agree on the rest. Let $D_i$ be $+1$, $-1$ or $0$ as the second classifier does better, worse or the same on example $i$. Use a paired t-test on the $D_i$. Why does the answer differ from part (b)? [5 marks]

$\overline{5}$

(d) You try 20 hyperparameter settings, none of which is really any better than the baseline, and test each at the 5% level. What is the probability that at least one looks significant? [3 marks]

3

(e) Check part (b) in Python by simulating the null hypothesis: two classifiers that are both right with probability $0.85$, each tested on its own 2000 examples. Repeat $100000$ times and find the fraction of trials in which the accuracies differ by at least $0.016$. [5 marks]

5

End of question 2

(a) $\frac{}{3}$ (b) $\frac{}{4}$ (c) $\frac{}{5}$ (d) $\frac{}{3}$ (e) $\frac{}{5}$ Total $\frac{}{20}$

**3**

(a) A disease affects 1% of people. A test detects it in 95% of people who have it, but is also positive for 5% of people who do not. Someone tests positive. What is the probability that they have the disease? [4 marks]

4

(b) A coin with unknown probability $p$ of a head is tossed 10 times, giving 7 heads. Starting from the uniform prior $\mathrm{Beta}(p \mid 1,1)$, what is the posterior? Find its mean and its most probable value, and compare with the frequency $7/10$. [4 marks]

$\overline{4}$

(c) Show that updating after each toss, using each posterior as the prior for the next toss, gives the same posterior as part (b), whatever the order of the heads and tails. [3 marks]

$\overline{3}$

(d) The number of cars passing in an hour is Poisson with rate $\mu$, and the prior is $\mathrm{Gamma}(\mu \mid 2,1)$. On three days we count 3, 5 and 4 cars. Using the update $a \to a+N$, $b \to b+1$ for each count, find the posterior and its mean and standard deviation ($a/b$ and $\sqrt{a}/b$). [4 marks]

$\overline{4}$

(e) In Python, simulate 100 tosses of a coin with $p = 0.7$ and update a $\mathrm{Beta}(1,1)$ prior after each toss. Print the posterior mean and standard deviation, $\sqrt{ab/((a+b)^2(a+b+1))}$, after 10, 30 and 100 tosses. [5 marks]

5

End of question 3

| (a) $\frac{}{4}$ | (b) $\frac{}{4}$ | (c) $\frac{}{3}$ | (d) $\frac{}{4}$ | (e) $\frac{}{5}$ | Total $\frac{}{20}$ |
|---|---|---|---|---|---|

**4** This question is harder and optional. When both the likelihood and the prior are normal, so is the posterior, and we only need to keep track of a mean and a variance (or a covariance matrix). A helpful way to read the prior is as *pseudo-data*: observations we pretend we have already seen.

(a) We measure an unknown quantity $\mu$ $n$ times, with independent errors: $X_i \sim \mathcal{N}(\mu,\sigma^2)$ with $\sigma$ known. Our prior is $\mu \sim \mathcal{N}(\mu_0,s_0^2)$. Write down the log of the posterior, up to a constant, and show that it is quadratic in $\mu$. Hence show the posterior is $\mathcal{N}(\mu_n,s_n^2)$ with

$$\frac{1}{s_n^2} = \frac{1}{s_0^2} + \frac{n}{\sigma^2}, \qquad \mu_n = s_n^2\left(\frac{\mu_0}{s_0^2} + \frac{1}{\sigma^2}\sum_{i=1}^{n} X_i\right).$$

(Hint: the log of $\mathcal{N}(\mu \mid m,s^2)$ is $-\mu^2/(2s^2) + \mu m/s^2 +$ a constant; match the coefficients of $\mu^2$ and $\mu$.) [5 marks]

5

(b) Let $n_0 = \sigma^2/s_0^2$ and $\bar{X} = \frac{1}{n}\sum_i X_i$. Show that $\mu_n = (n_0\mu_0 + n\bar{X})/(n_0 + n)$ and $s_n^2 = \sigma^2/(n_0+n)$. Interpret $n_0$. What happens when $s_0 \to \infty$, and when $n \to \infty$? [4 marks]

4

(c) Take $\mu_0 = 0$, $s_0 = 2$, $\sigma = 1$ and measurements $1.2, 0.8, 1.5, 0.9$. Find $n_0$, $\mu_n$ and $s_n$. [3 marks]

3

(d) Now consider linear regression, $y_k = \boldsymbol{w}^\mathsf{T}\boldsymbol{\phi}_k + \epsilon_k$ with $\epsilon_k \sim \mathcal{N}(0,\sigma^2)$, where the feature vectors $\boldsymbol{\phi}_k$ are the rows of a matrix $\boldsymbol{\Phi}$. Take the prior $\boldsymbol{w} \sim \mathcal{N}(\mathbf{0},(\sigma^2/\eta)\mathbf{I})$: the weights are probably small. Using the multivariate version of the hint, $\mathrm{e}^{-\frac{1}{2}\boldsymbol{w}^\mathsf{T}\mathbf{A}\boldsymbol{w}+\boldsymbol{b}^\mathsf{T}\boldsymbol{w}} \propto \mathcal{N}(\boldsymbol{w} \mid \mathbf{A}^{-1}\boldsymbol{b},\mathbf{A}^{-1})$ (we show this in the lecture on Gaussian integrals), show that the posterior has precision $\mathbf{A} = (\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})/\sigma^2$ and mean $(\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{\Phi} + \eta\mathbf{I})^{-1}\boldsymbol{\Phi}^\mathsf{T}\boldsymbol{y}$. What do you recognise? [6 marks]

6

(e) Fit a line $y = w_1 + w_2 x$ ($\boldsymbol{\phi}_k = (1, x_k)^\mathsf{T}$) to the points $(0, 1.1)$, $(1, 2.9)$, $(2, 5.2)$, $(3, 6.8)$ with $\sigma = 1$ and $\eta = 0.5$. In Python, update the posterior one point at a time, by adding to the precision and to $\boldsymbol{b}$, and print the posterior mean and the standard deviations of $w_1$ and $w_2$ after each point. Check the final mean against ridge regression. [7 marks]

$\frac{}{7}$

End of question 4

(a) $\frac{}{5}$ (b) $\frac{}{4}$ (c) $\frac{}{3}$ (d) $\frac{}{6}$ (e) $\frac{}{7}$ Total $\frac{}{25}$